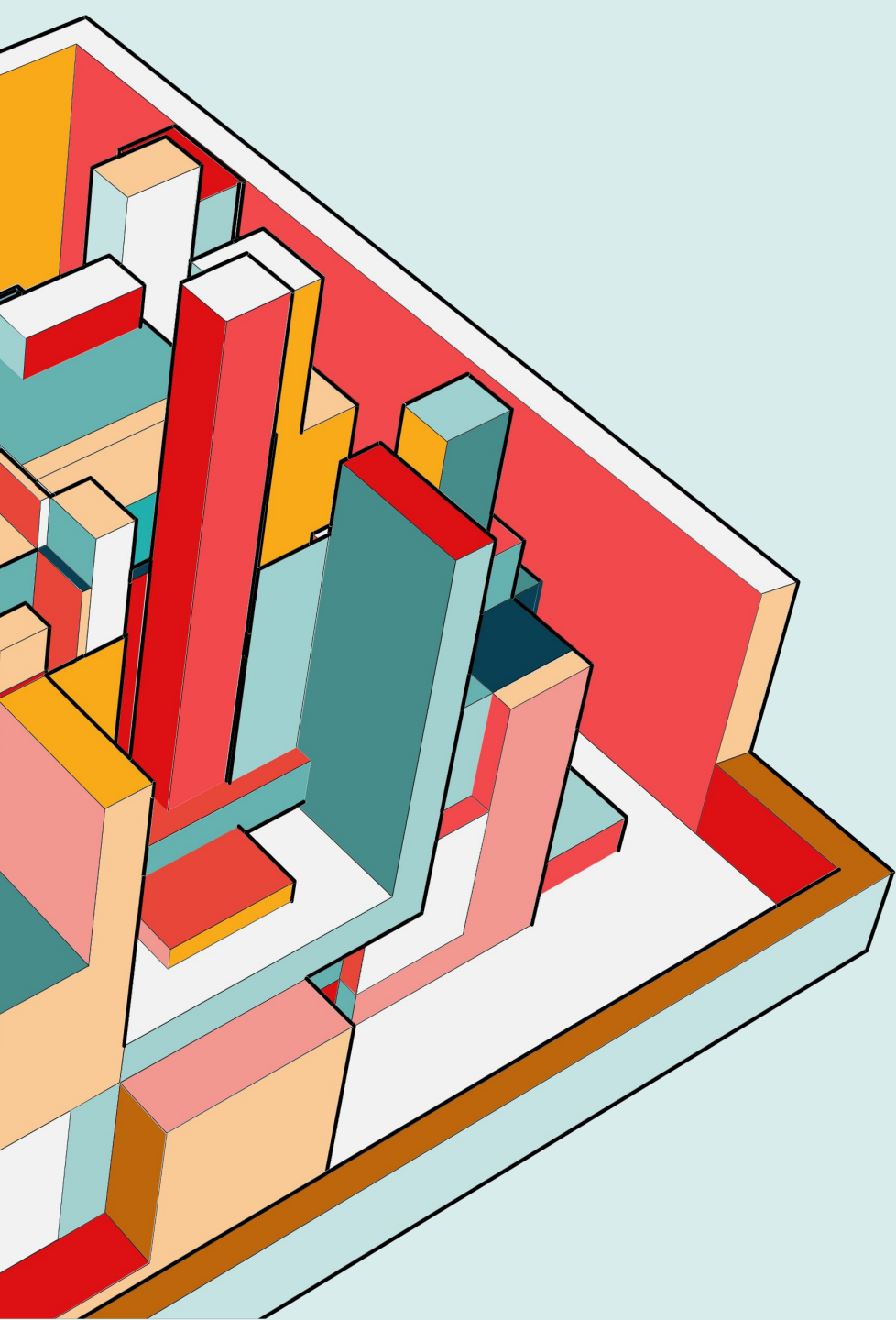




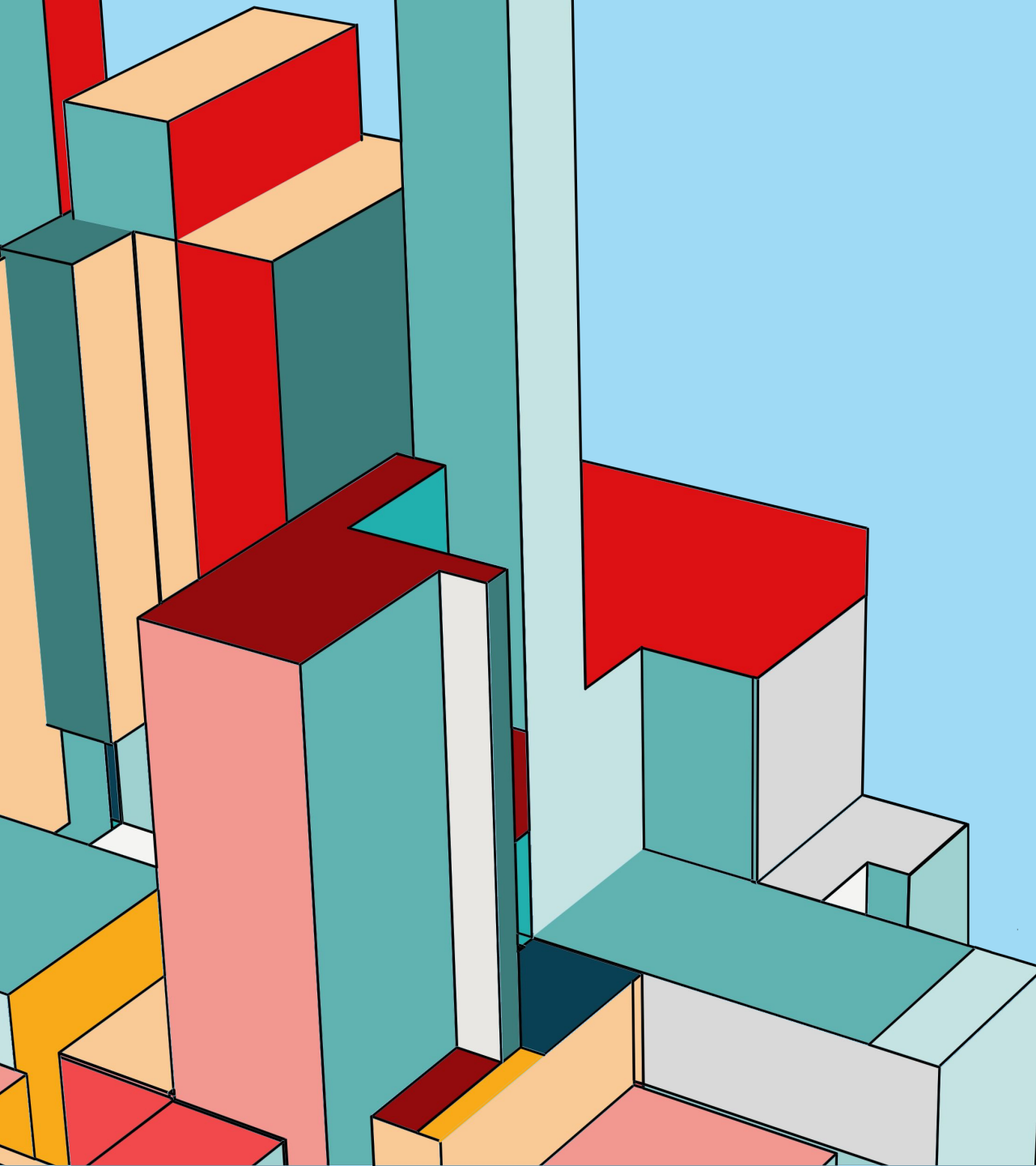
Methods 3 - Multilevel Statistical Modelling & Machine Learning

Classroom Instruction

WEEK 39

AGENDA

1. Covering last week
 - a. multilevel modelling essentials
 - b. assignment_0
2. Today
 - a. assignment_1 released?
3. Next week
 - a. assignment_1
 - b. W39 readings + lecture
 - i. logistic regression
 - c. W40 reading + lecture



Covering last week

Covering Week 38

varying-intercept model: $y_i = \alpha_{j[i]} + \beta x_i + \epsilon_i$.

Another option, shown in Figure 11.1b, is to let the slope vary with constant intercept:

varying-slope model: $y_i = \alpha + \beta_{j[i]} x_i + \epsilon_i$.

Finally, Figure 11.1c shows a model in which both the intercept and the slope vary by group:

varying-intercept, varying-slope model: $y_i = \alpha_{j[i]} + \beta_{j[i]} x_i + \epsilon_i$.

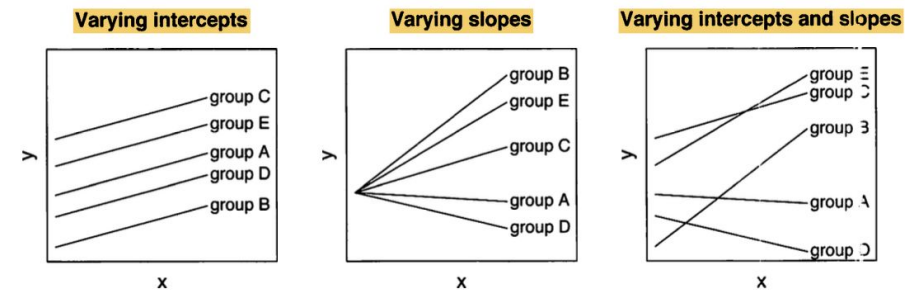


Figure 11.1 Linear regression models with (a) varying intercepts ($y = \alpha_j + \beta x$), (b) varying slopes ($y = \alpha + \beta_j x$), and (c) both ($y = \alpha_j + \beta_j x$). The varying intercepts correspond to group indicators as regression predictors, and the varying slopes represent interactions between x and the group indicators.

Covering Week 38

$$Y = X\beta + Zu + \epsilon$$

Y : a column vector with n observations (known)

X : the first level design matrix (known), size: $n \times p$

β : unknown $p \times 1$ column vector of the first level coefficients

Z : $n \times q$ design matrix (known) for the q random effects

u : unknown $q \times 1$ column vector of the second-level coefficients: $N(0, G)$

G : the covariance matrix of the second-level effects (unknown): $q \times q$

ϵ : $N \times 1$ column vector (unknown) of the residuals: $N(0, \sigma^2 I)$

Covering Week 38

$$Y = X\beta + Zu + \epsilon$$

Now we have two design matrices,
 X and Z , but no analytical solution.
(What does it mean and entail that
we have no analytical solutions?)

Covering Week 38

Variance-covariance matrix

$$\mathbf{u} \sim N(\mathbf{0}, \mathbf{\Sigma}) \quad (3)$$

$$\mathbf{\Sigma}_{\theta} = \sigma^2 \mathbf{\Lambda}_{\theta} \mathbf{\Lambda}_{\theta}^T \quad (4)$$

$$\mathbf{G} = \mathbf{\Lambda}_{\theta} \mathbf{\Lambda}_{\theta}^T$$

of two vector-valued random variables: $\mathcal{Y}$, the response, and $\mathcal{B}$, the vector of random effects. In a linear model the distribution of $\mathcal{Y}$ is multivariate normal,

$$\mathcal{Y} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta} + \mathbf{o}, \sigma^2 \mathbf{W}^{-1}), \quad (1)$$

where n is the dimension of the response vector, $\mathbf{W}$ is a diagonal matrix of known prior weights, $\boldsymbol{\beta}$ is a p -dimensional coefficient vector, $\mathbf{X}$ is an $n \times p$ model matrix, and $\mathbf{o}$ is a vector of known prior offset terms. The parameters of the model are the coefficients $\boldsymbol{\beta}$ and the scale parameter σ .

In a linear mixed model it is the *conditional* distribution of $\mathcal{Y}$ given $\mathcal{B} = \mathbf{b}$ that has such a form,

$$(\mathcal{Y}|\mathcal{B} = \mathbf{b}) \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{b} + \mathbf{o}, \sigma^2 \mathbf{W}^{-1}), \quad (2)$$

where $\mathbf{Z}$ is the $n \times q$ model matrix for the q -dimensional vector-valued random-effects variable, $\mathcal{B}$, whose value we are fixing at $\mathbf{b}$. The unconditional distribution of $\mathcal{B}$ is also multivariate normal with mean zero and a parameterized $q \times q$ variance-covariance matrix, $\mathbf{\Sigma}$,

$$\mathcal{B} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}). \quad (3)$$

As a variance-covariance matrix, $\mathbf{\Sigma}$ must be positive semidefinite. It is convenient to express the model in terms of a *relative covariance factor*, $\mathbf{\Lambda}_{\theta}$, which is a $q \times q$ matrix, depending on the *variance-component parameter*, θ , and generating the symmetric $q \times q$ variance-covariance matrix, $\mathbf{\Sigma}$, according to

$$\mathbf{\Sigma}_{\theta} = \sigma^2 \mathbf{\Lambda}_{\theta} \mathbf{\Lambda}_{\theta}^T, \quad (4)$$

where σ is the same scale factor as in the conditional distribution (2).

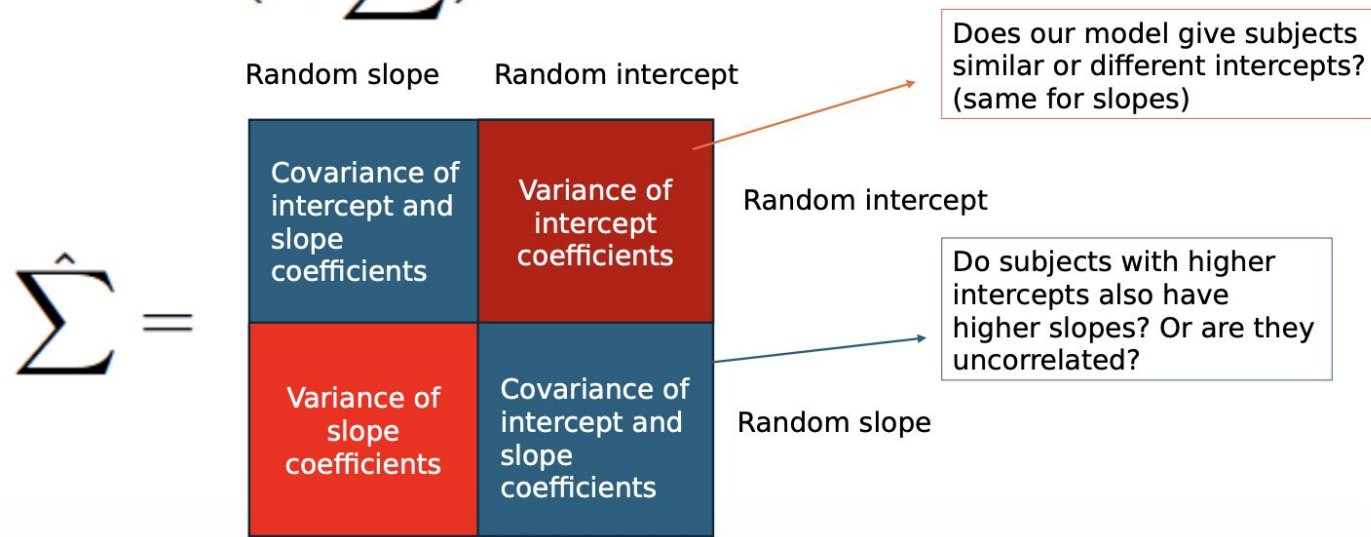
Although Equations 2, 3, and 4 fully describe the class of linear mixed models that **lme4** can fit, this terse description hides many important details. Before moving on to these details, we make a few observations:

- This formulation of linear mixed models allows for a relatively compact expression for the profiled log-likelihood of θ (Section 3.4, Equation 34).
- The matrices associated with random effects, $\mathbf{Z}$ and $\mathbf{\Lambda}_{\theta}$, typically have a sparse structure with a sparsity pattern that encodes various model assumptions. Sections 2.3 and 3.7 provide details on these structures, and how to represent them efficiently.
- The interface provided by **lme4**'s **lmer** function is slightly less general than the model described by Equations 2, 3, and 4. To take advantage of the entire range of possibilities, one may use the modular functions (Sections 1.3 and Appendix A) or explore the experimental **flexLambda** branch of **lme4** on Github.

What is this weird E? It's a matrix - not a sum!

Notation clarification

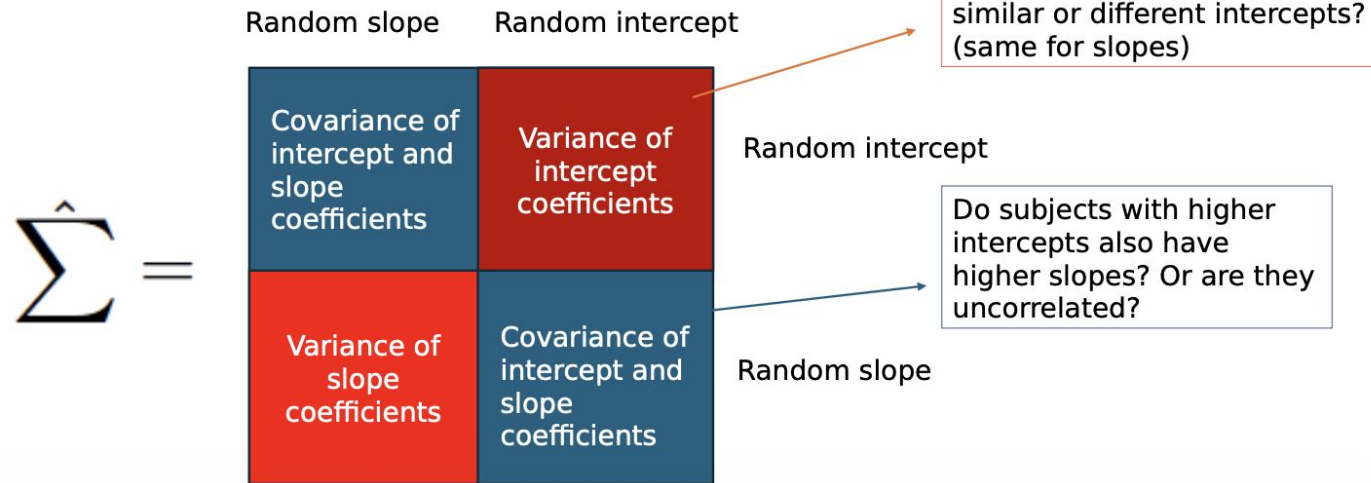
$$u \sim N(0, \hat{\Sigma})$$



What is this weird E? It's a matrix - not a sum!

Notation clarification

$$u \sim N(0, \hat{\Sigma})$$



```
# wk38
...{r}

data("sleepstudy")
df <- sleepstudy

mlm1 <- lmer(Reaction ~ Days + (Days | Subject), data = df)
VarCorr(mlm1)
# ranef(mlm1)$Subject[1,2]
# ranef(mlm1)$Subject
# fixef(mlm1)
...

Groups    Name           Std.Dev.   Corr
Subject   (Intercept)   24.7407
            Days           5.9221   0.066
Residual                   25.5918
```

$$u \sim N(0, \Sigma)$$
$$\hat{\Sigma}_{(1:2, 1:2)} = \begin{pmatrix} 24.741^2 & 9.61 \\ 9.61 & 5.922^2 \end{pmatrix}$$

What is this weird E? It's a matrix - not a sum!

Variance-covariance matrix

$$u \sim N(0, \Sigma) \quad (3)$$

$$\Sigma_{\theta} = \sigma^2 \Lambda_{\theta} \Lambda_{\theta}^T \quad (4)$$

$$\mathbf{G} = \Lambda_{\theta} \Lambda_{\theta}^T$$

```
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```

Groups	Name	Std.Dev.	Corr
Subject	(Intercept)	24.7407	
	Days	5.9221	0.066
Residual		25.5918	

$$u \sim N(0, \Sigma)$$

$$\hat{\Sigma}_{(1:2, 1:2)} = \begin{bmatrix} 24.741^2 & 9.61 \\ 9.61 & 5.922^2 \end{bmatrix}$$

$$\mathbf{G}_{(1:2, 1:2)} = \begin{bmatrix} 0.93 & 0.015 \\ 0.015 & 0.054 \end{bmatrix}$$

Covering Week 38

The weird E is a variance-covariance matrix with a diagonal of variance-covariance matrices

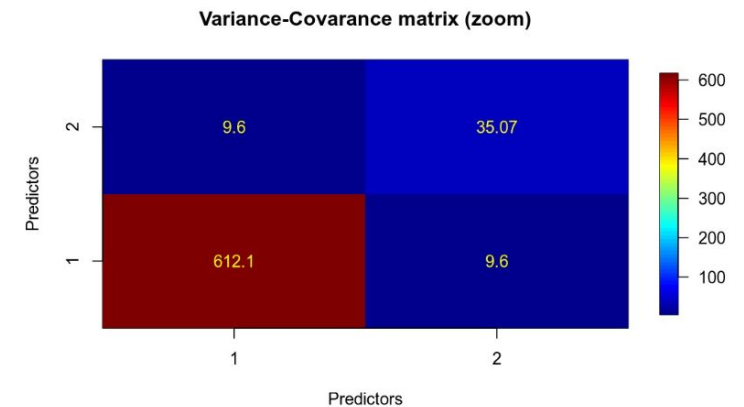
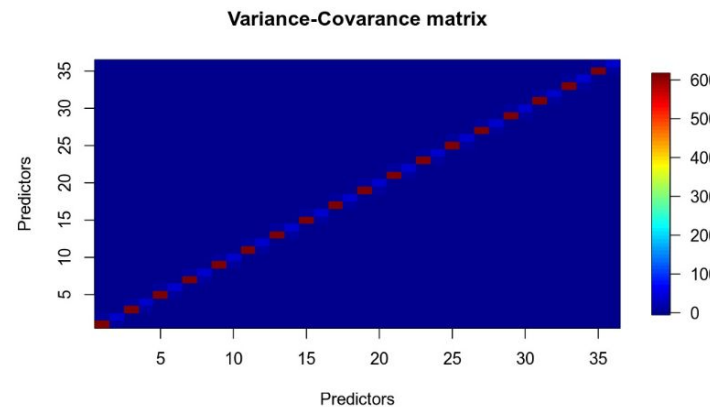
In the sleep study case there is the SAME variance-covariance matrix for each subject ($n = 18$) along the diagonal.

Each of those matrices has variance for each predictor (random slope and random intercept) and covariance of random intercept and random slope.

2 predictors · 18 subject = 36

It is used for the partial-pooling estimates of y !

$$\hat{\Sigma}_{\theta}$$



Q: What would the plot look like if there was no covariance between predictors?

$$Corr = \frac{9.6}{\sqrt{(612.1 * 35.07)}} = 0.07$$

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Covering Week 38 - What the helly?

$$Y = X\beta + Zu + \epsilon$$

Henderson's equation

$$\begin{pmatrix} X^T X & X^T Z \\ Z^T Z & Z^T Z G^{-1} \end{pmatrix} \begin{pmatrix} \hat{\beta} \\ \hat{u} \end{pmatrix} = \begin{pmatrix} X^T y \\ Z^T y \end{pmatrix}$$

Maximise this

$$l_{ML}(\hat{\beta}, \hat{u}, \hat{\sigma}^2 | y) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \log |G| - \frac{1}{2} \log |G^{-1} + Z^T Z| - \frac{n}{2} \log\left(\frac{r^2}{n}\right) - \frac{n}{2}$$

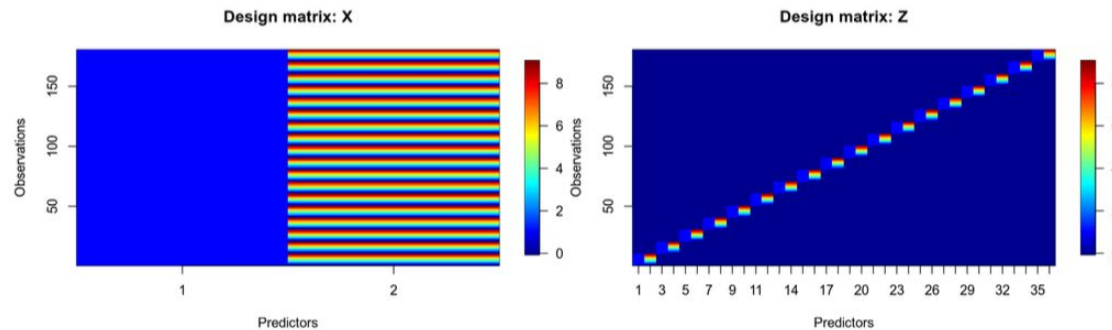
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27

Covering Week 38

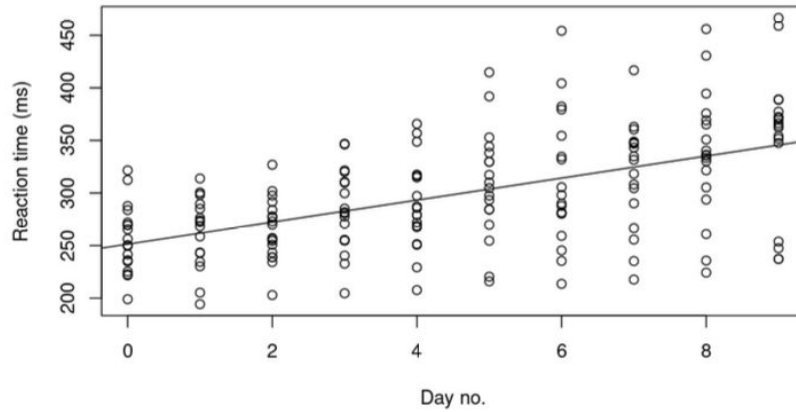
$$Y = \underbrace{X\beta}_{\text{Fixed effects}} + \underbrace{Zu}_{\text{Random effects}} + \epsilon$$

Level 1 Level 2

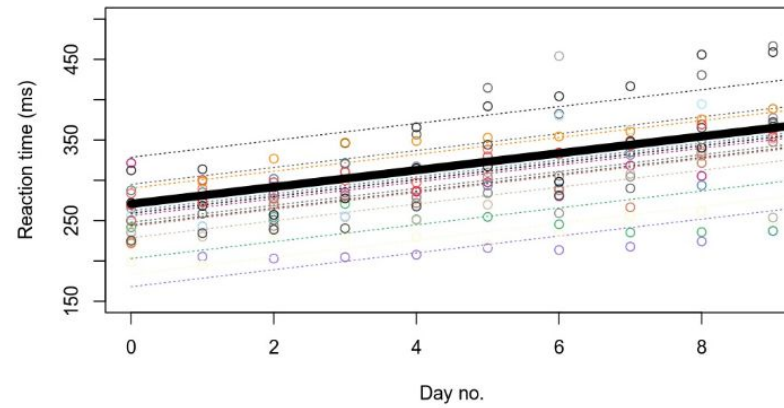


Covering Week 38

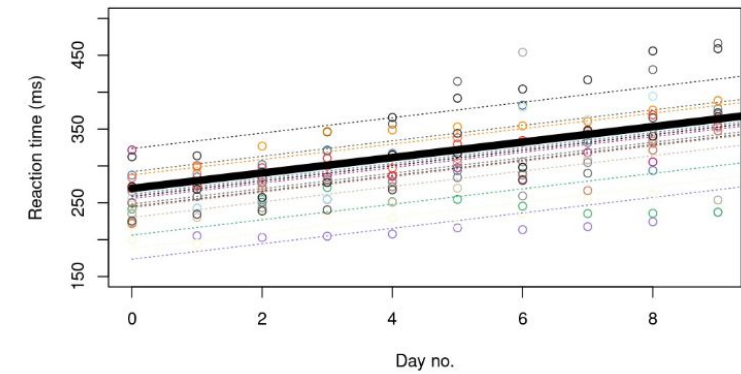
Pooled linear regression



No-pooling model



Linear regression with subject-level intercepts



Covering Week 38

$$\alpha_j \approx \frac{\frac{n_j}{\sigma_y^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} (\bar{y}_j - \beta \bar{x}_j) + \frac{\frac{1}{\sigma_\alpha^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} \mu_\alpha$$

Q's:

What happens to the estimated α_j ,
when respectively n_j , σ_y , or σ_α :

- 1) increases?
- 2) decreases?
- 3) is 0?
- 4) goes towards infinity?

α_j : partially pooled response

$(\bar{y}_j - \beta \bar{x}_j)$: subject estimate of mean

μ_α : group estimate of mean

σ_y^2 : within-group variance

σ_α^2 : between-group variance

$\frac{\frac{n_j}{\sigma_y^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}}$: proportion assigned to subject estimate

$\frac{\frac{1}{\sigma_\alpha^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}}$: proportion assigned to group estimate

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$\frac{\frac{n_j}{\sigma_y^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} + \frac{\frac{1}{\sigma_\alpha^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} = 1$$

Covering Week 38

$$\alpha_j \approx \frac{\frac{n_j}{\sigma_y^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} (\bar{y}_j - \beta \bar{x}_j) + \frac{\frac{1}{\sigma_\alpha^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} \mu_\alpha$$

The weighted average (12.1) reflects the relative amount of information available about the individual county, on one hand, and the average of all the counties, on the other:

- Averages from counties with smaller sample sizes carry less information, and the weighting pulls the multilevel estimates closer to the overall state average. In the limit, if $n_j = 0$, the multilevel estimate is simply the overall average, $\bar{y}_{\text{all}}$.
- Averages from counties with larger sample sizes carry more information, and the corresponding multilevel estimates are close to the county averages. In the limit as $n_j \rightarrow \infty$, the multilevel estimate is simply the county average, $\bar{y}_j$.
- In intermediate cases, the multilevel estimate lies between the two extremes.

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Covering Week 38

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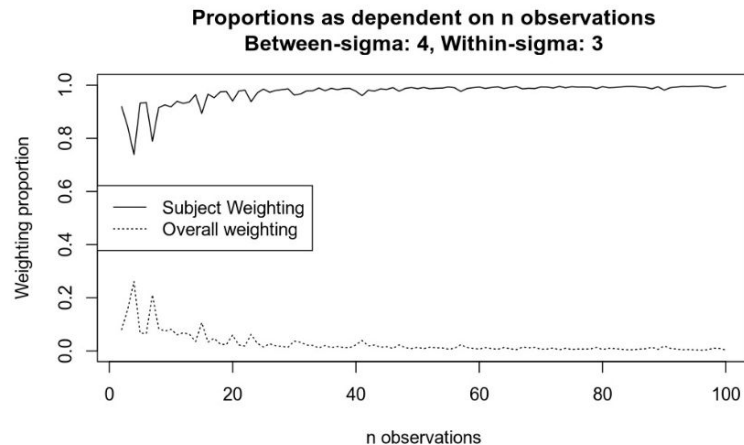
σ_y^2 : within-group variance

σ_α^2 : between-group variance

$$\alpha_j \approx \frac{\frac{n_j}{\sigma_y^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} (\bar{y}_j - \beta \bar{x}_j) + \frac{\frac{1}{\sigma_\alpha^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} \mu_\alpha$$

When $\sigma_\alpha > \sigma_y$

100 SIMULATIONS



$$\frac{\frac{n_j}{\sigma_y^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}}: \text{proportion assigned to subject estimate}$$

$$\frac{\frac{1}{\sigma_\alpha^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}}: \text{proportion assigned to group estimate}$$

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

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Covering Week 38

α_j : partially pooled response

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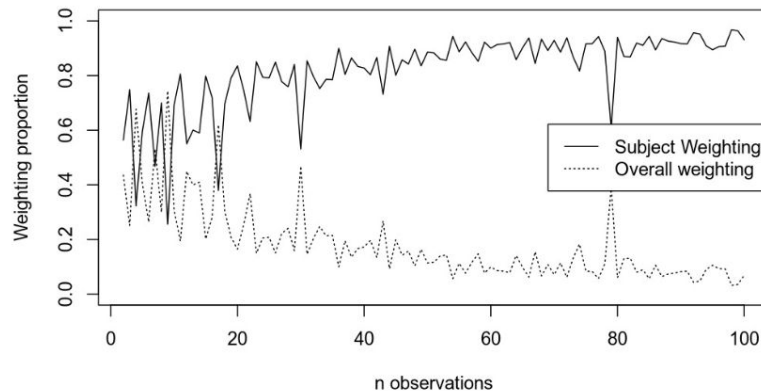
σ_α^2 : between-group variance

$$\alpha_j \approx \frac{\frac{n_j}{\sigma_y^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} (\bar{y}_j - \beta \bar{x}_j) + \frac{\frac{1}{\sigma_\alpha^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} \mu_\alpha$$

When $\sigma_\alpha < \sigma_y$

100 SIMULATIONS

Proportions as dependent on n observations
Between-sigma: 4, Within-sigma: 10



$$\frac{\frac{n_j}{\sigma_y^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}}: \text{proportion assigned to subject estimate}$$

$$\frac{\frac{1}{\sigma_\alpha^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}}: \text{proportion assigned to group estimate}$$

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

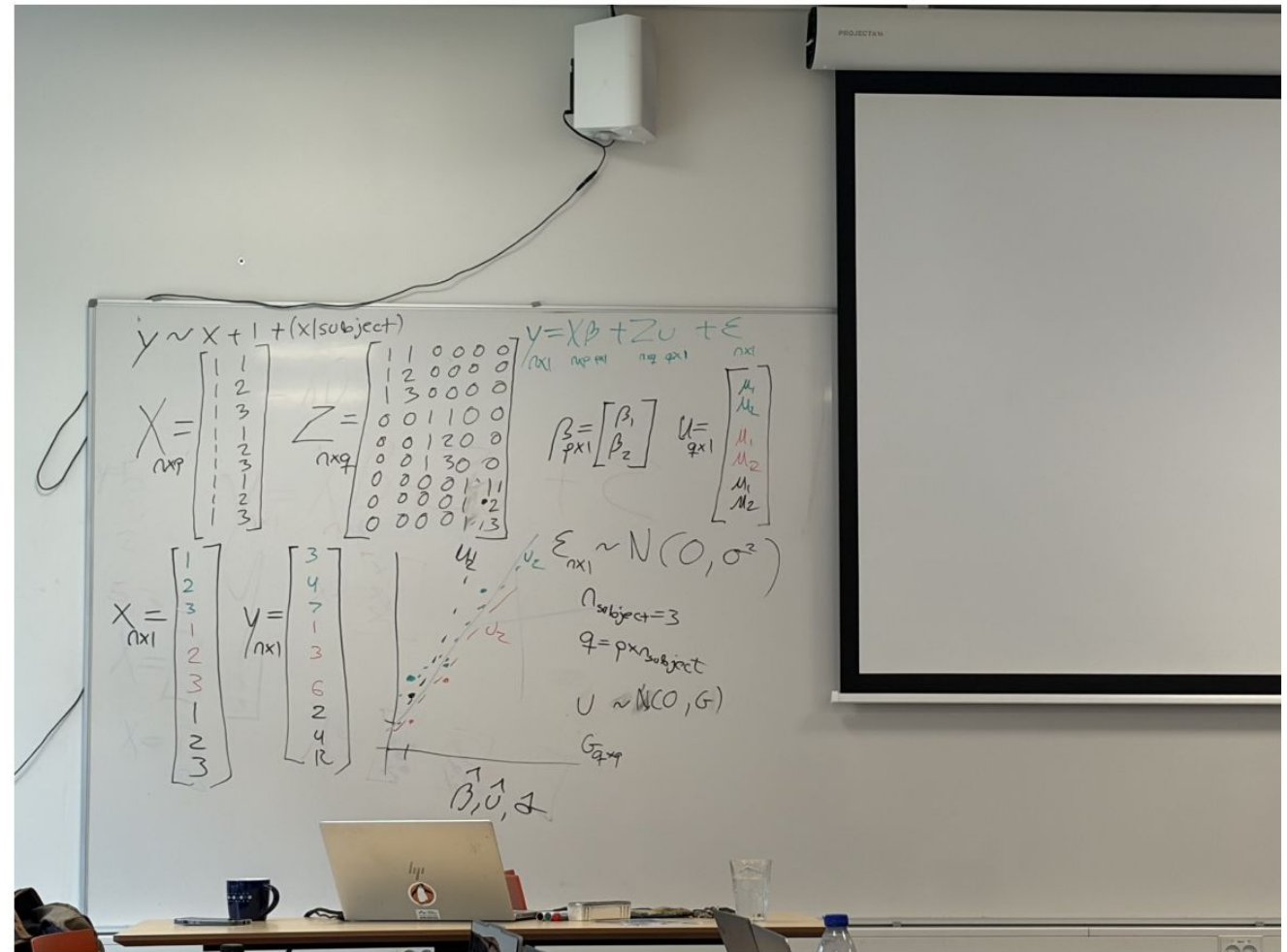
$$\frac{\frac{n_j}{\sigma_y^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} + \frac{\frac{1}{\sigma_\alpha^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} = 1$$

Covering Week 38

Should we try to make our own example?

Perchance with lizards and length growth?

Or are you OK for now?



$x_{n \times 1}$ is the independent variable from three different subjects, colour coded.

$y_{n \times 1}$ is the dependent variable, measured from the same subjects, colour coded.

$X_{n \times p}$ is the design matrix for the general effects across participants (first level effects) (intercept and slope)

$Z_{q \times p}$ is the design matrix for the effects for each participant (second level effects) (intercept and slope for each subject)

$\beta_{p \times 1}$ are the coefficients for the two parameters of X

$u_{q \times 1}$ are the coefficients for the six parameters of Z ($q = n_{subjects} \times p$)

$\epsilon_{n \times 1}$ are the residuals, assumed to normally distributed with a mean of 0 and a variance of σ^2

u is normally distributed with a mean of 0 and a variance-covariance structure $G_{q \times q}$

From Lau's class in W37

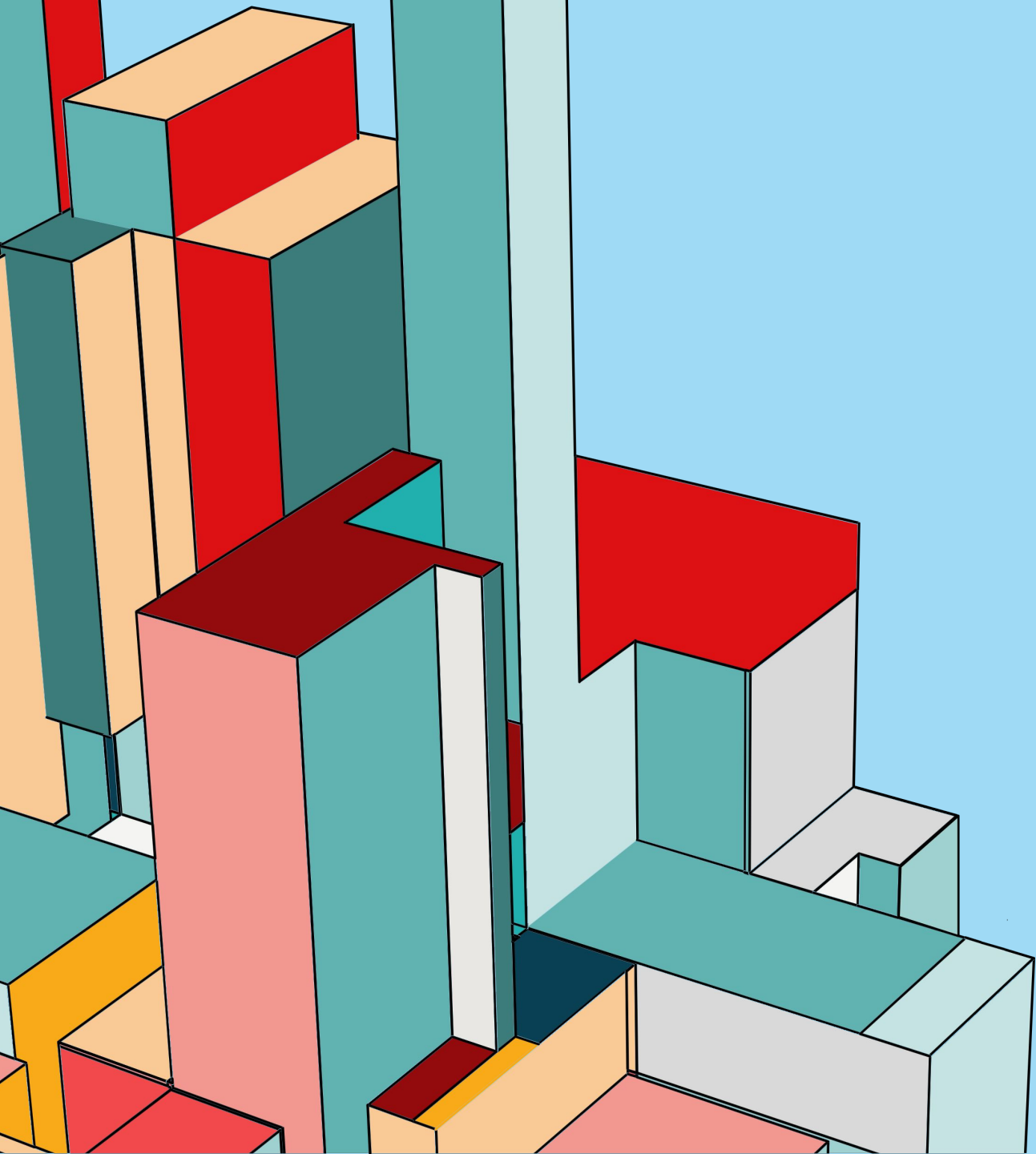
For visual explanation of Multilevel Modelling

https://www.youtube.com/watch?v=ol1_SV1Rpfc

Assignment_0

Formula	Alternative	Meaning
$(1 \mid g)$	$1 + (1 \mid g)$	Random intercept with fixed mean.
$0 + \text{offset}(o) + (1 \mid g)$	$-1 + \text{offset}(o) + (1 \mid g)$	Random intercept with <i>a priori</i> means.
$(1 \mid g1/g2)$	$(1 \mid g1) + (1 \mid g1:g2)$	Intercept varying among $g1$ and $g2$ within $g1$.
$(1 \mid g1) + (1 \mid g2)$	$1 + (1 \mid g1) + (1 \mid g2)$	Intercept varying among $g1$ and $g2$.
$x + (x \mid g)$	$1 + x + (1 + x \mid g)$	Correlated random intercept and slope.
$x + (x \parallel g)$	$1 + x + (1 \mid g) + (0 + x \mid g)$	Uncorrelated random intercept and slope.

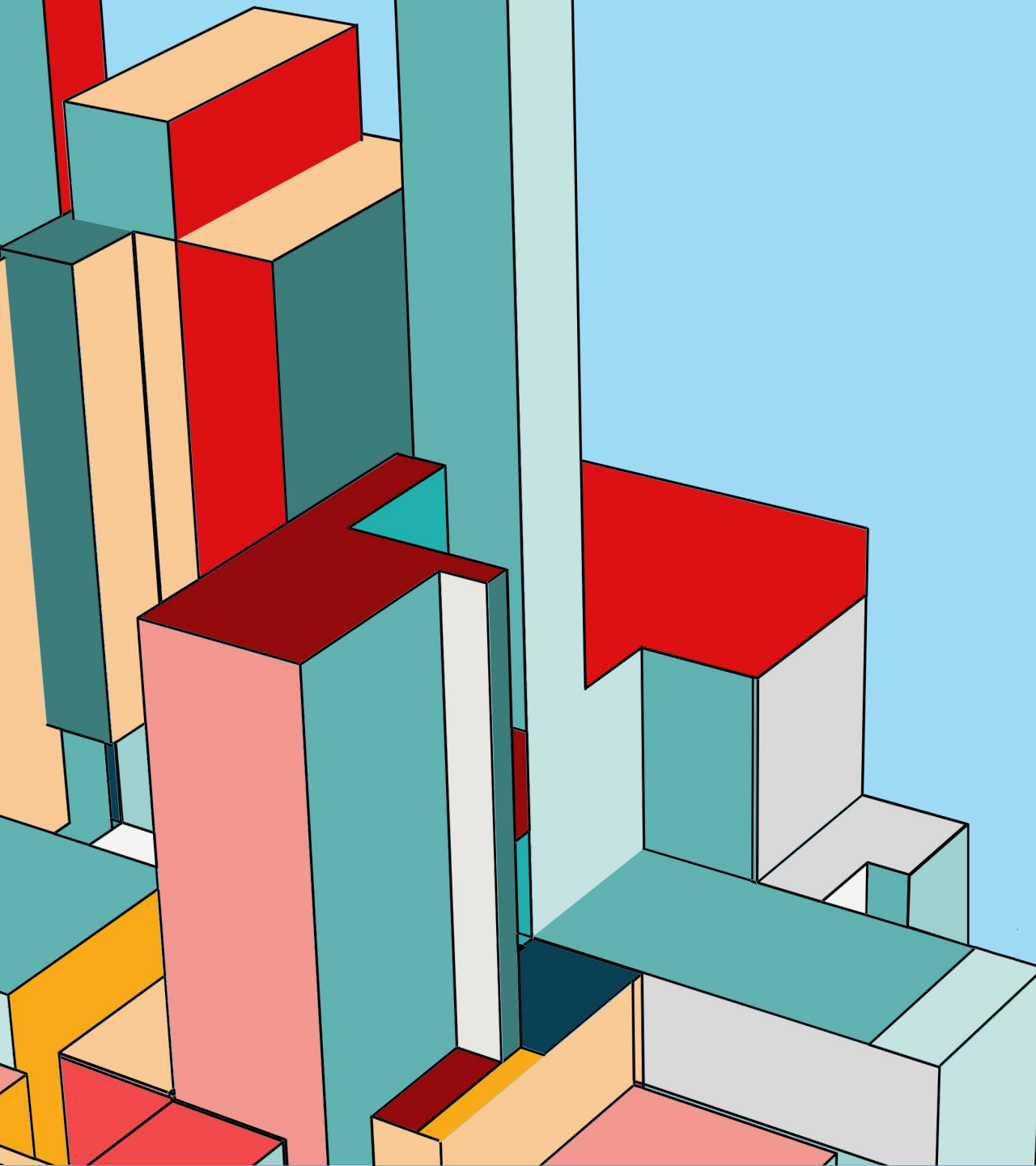
Table 2: Examples of the right-hand-sides of mixed-effects model formulas. The names of grouping factors are denoted g , $g1$, and $g2$, and covariates and *a priori* known offsets as x and o .



Today

assignment_1 has

change to Lau's slides.



For Next Week

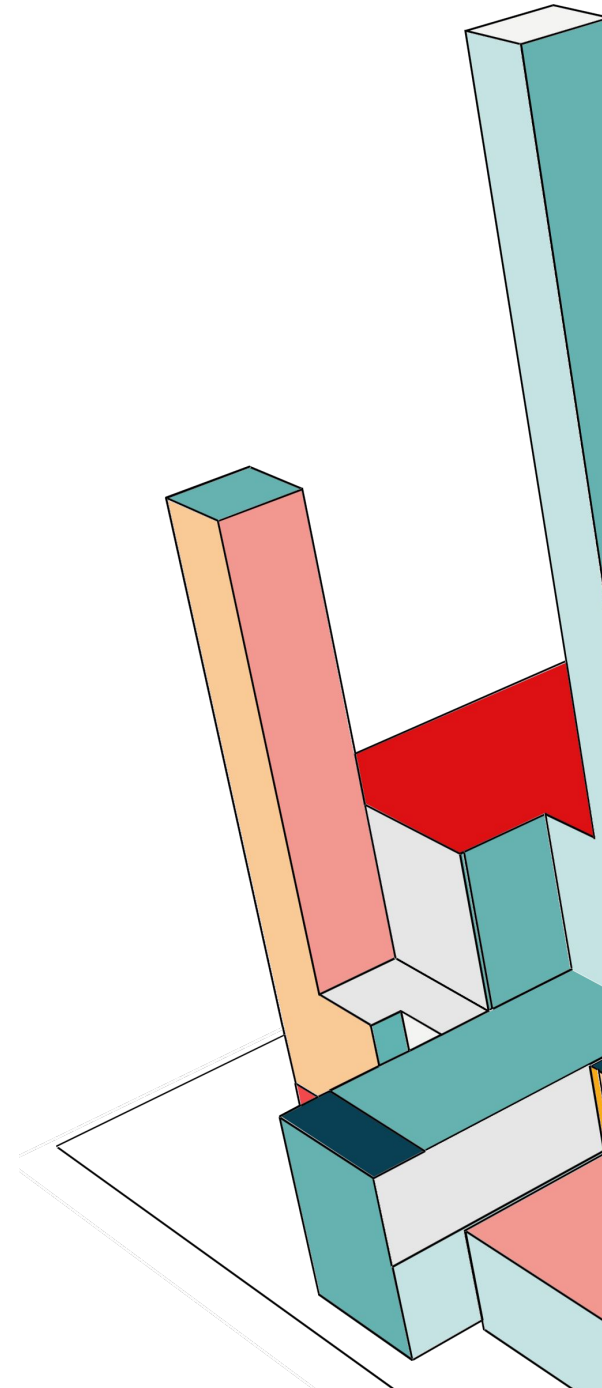
FOR NEXT WEEK

Covering this week's readings and lecture on logistic regression.

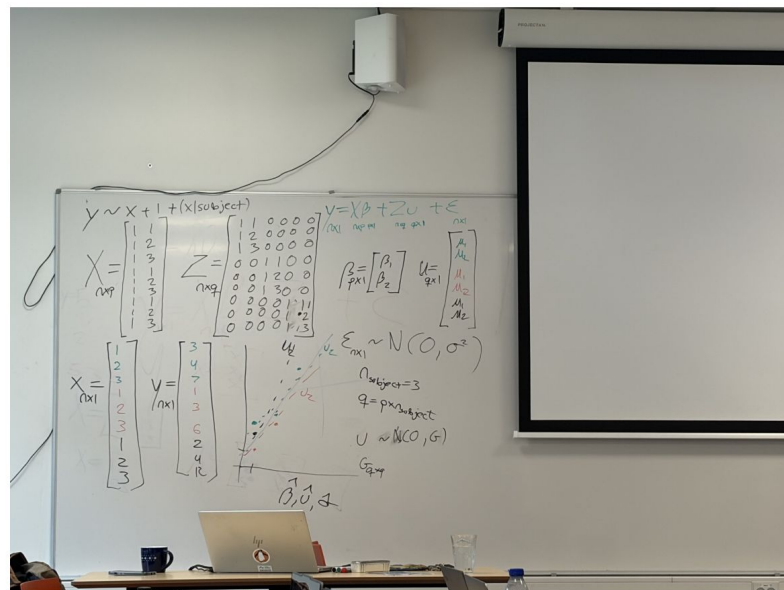
+

assignment_1

Any other wishes?



SEE YOU NEXT WEEK!



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